Drug-Specific Regression and SHAP Analysis of Cancer Drug Screening Data

Machine Learning: Assignment 4

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Abstract. This report predicts how sensitive 963 cancer cell lines are to 13 anti-cancer drugs, using the expression of 2,528 genes. For every drug, six tree-based regressors (Decision Tree, Random Forest, Gradient Boosting Machine, XGBoost, LightGBM and CatBoost) are evaluated with 5-fold cross-validation using MAE, MSE, RMSE and R^2. A boosting model is the best regressor for every drug, with R^2 between 0.15 (Pictilisib) and 0.60 (Nutlin-3a). SHAP analysis of the best models identifies the ten most influential genes per drug, and many of them, such as BCL2 for Navitoclax, MDM2 for Nutlin-3a and FGFR2 for PD173074, are the known targets or pathway markers of the drugs.

1. Introduction

Cancer cell lines respond very differently to the same drug. Part of this variation is explained by the genes each cell line expresses, so gene expression can be used to predict drug sensitivity and to discover biomarkers that indicate which tumours are likely to respond. This report addresses three tasks:

- **Task 1, understanding the data:** show how the samples are distributed over the 13 drugs.
- **Task 2, drug-specific regression:** for each drug, train six tree-based regressors and identify the best one using MAE, MSE, RMSE and R^2.
- **Task 3, explainability:** apply SHAP to the best regressor of each drug and identify its ten most important genes.

Section 2 describes the algorithms, approaches and tools. Section 3 describes the data and the experimental setup, Section 4 presents the results of the three tasks, and Section 5 concludes.

2. Algorithms, Approaches and Tools

For each method, this section describes (a) what it is or does, (b) how it works, and (c) how it is applied in this assignment.

2.1 Decision Tree Regressor

(a) What it is. A decision tree is a supervised model that predicts a numeric target by asking a sequence of yes/no questions about the features. It is the basic building block of every other model in this study.

(b) How it works. Starting from all training samples, the tree searches every feature and threshold for the split (for example, *expression of BCL2* ≤ 5.3) that most reduces the squared error of the target in the two resulting groups. It then repeats the search inside each group, recursively, until a stopping rule is met. Each leaf predicts the mean target of the training samples that reach it. A single deep tree has low bias but high variance: it can memorise noise in the training data.

(c) Application. DecisionTreeRegressor from scikit-learn, with *max_depth = 5* and *min_samples_leaf = 10* to limit overfitting. It serves as the baseline that the ensemble methods are compared against.

2.2 Random Forest

(a) What it is. A random forest is an ensemble of many decision trees whose predictions are averaged. It is a *bagging* (bootstrap aggregating) method.

(b) How it works. Each tree is trained on a bootstrap sample (rows drawn with replacement), and at each split only a random subset of the features is considered. The randomness makes the trees different from each other, so their errors are partly independent; averaging them cancels much of the variance of individual trees while keeping their low bias. The trees are trained independently, so they can be built in parallel.

(c) Application. RandomForestRegressor from scikit-learn with 300 trees, *max_features = "sqrt"* (about 50 of the 2,528 genes per split) and *min_samples_leaf = 5*.

2.3 Gradient Boosting Machine (GBM)

(a) What it is. Gradient boosting is an ensemble method that builds trees *sequentially*, each new tree correcting the errors of the ensemble so far.

(b) How it works. The model starts from a constant prediction (the mean). At each round it computes the residuals, which for squared-error loss are the negative gradient of the loss, and fits a small tree to them. The tree's prediction, scaled by a learning rate, is added to the ensemble. Many shallow trees and a small learning rate give an accurate model that mainly reduces bias. With *subsample* < 1, each tree sees a random fraction of the rows (stochastic gradient boosting), which adds regularisation.

(c) Application. `GradientBoostingRegressor` from scikit-learn with 200 trees, *learning_rate* $= 0.05$, *max_depth* $= 3$ and *subsample* $= 0.8$.

2.4 XGBoost

(a) What it is. XGBoost (eXtreme Gradient Boosting) is an optimised and regularised implementation of gradient boosting.

(b) How it works. Like GBM, it adds trees sequentially, but it uses both the first and second derivatives of the loss (a Newton step) to choose splits and leaf values, and it adds L1/L2 penalties on the leaf weights to the objective. It also supports row and column subsampling, handles missing values natively, and parallelises the split search, which makes it fast on wide data.

(c) Application. `XGBRegressor` with 300 trees, *learning_rate* $= 0.05$, *max_depth* $= 4$, *subsample* $= 0.8$ and *colsample_bytree* $= 0.5$ (each tree sees a random half of the genes).

2.5 LightGBM

(a) What it is. LightGBM (Light Gradient Boosting Machine), developed by Microsoft, is a gradient boosting framework designed for speed and low memory use on large data sets.

(b) How it works. It bins continuous features into histograms, so split search works on a few hundred bins instead of every distinct value. It grows trees *leaf-wise*: it always splits the leaf with the largest loss reduction, rather than growing level by level. This gives more accurate trees for the same number of leaves, so tree size is controlled with *num_leaves* instead of depth.

(c) Application. `LGBMRegressor` with 300 trees, *learning_rate* $= 0.05$, *num_leaves* $= 15$, *subsample* $= 0.8$ and *colsample_bytree* $= 0.5$.

2.6 CatBoost

(a) What it is. CatBoost (Categorical Boosting), developed by Yandex, is a gradient boosting library known for strong results with little tuning.

(b) How it works. It builds *symmetric (oblivious) trees*, in which every node at the same depth uses the same split. These trees are fast to evaluate and act as a strong regulariser. It also uses *ordered boosting*: the residual for each sample is computed by a model that has not seen that sample, which reduces the target leakage and overfitting of standard boosting. Its native handling of categorical features is not needed here, because all features are numeric.

(c) Application. `CatBoostRegressor` with 300 iterations, *learning_rate* $= 0.05$ and *depth* $= 4$.

2.7 K-Fold Cross-Validation

(a) What it is. K-fold cross-validation estimates how well a model will perform on unseen data.

(b) How it works. The data is shuffled and split into K equal folds. The model is trained K times, each time on K-1 folds, and tested on the remaining fold. Every sample is used for testing exactly once, and the K test scores are averaged. This is more reliable than a single train/test split, especially with fewer than 1,000 samples per drug.

(c) Application. `KFold(n_splits=5, shuffle=True, random_state=42)`. The same folds are used for all six models of a drug, so the models are compared on exactly the same data. Each drug therefore needs 6 x 5 = 30 model fits, and the whole study needs 390.

2.8 Evaluation Metrics: MAE, MSE, RMSE and R^2

(a) What it is. Four standard regression metrics compare the predicted LN_IC50 values with the measured ones.

(b) How it works. MAE (mean absolute error) is the average of $|y - \hat{y}|$ and weighs all errors equally. **MSE** (mean squared error) is the average of $(y - \hat{y})^2$ and penalises large errors more heavily. **RMSE** is the square root of MSE, which puts it back in the units of the target. $\mathbf{R^2} = 1 - \text{MSE} / \text{Var}(y)$ is the share of the target's variance explained by the model: 1 is perfect, 0 is no better than predicting the mean, and negative values are worse than the mean.

(c) Application. All four metrics are computed on each held-out fold and averaged over the 5 folds. The best model per drug is the one with the lowest RMSE, for the reasons explained in Section 3.3.

2.9 SHAP (SHapley Additive exPlanations)

(a) What it is. SHAP is a method that explains individual predictions of any machine learning model by assigning each feature a contribution to the prediction.

(b) How it works. It is based on Shapley values from cooperative game theory. A feature's SHAP value for one sample is its average marginal contribution to the prediction over all possible orders in which the features could be added. The SHAP values of a sample add up exactly to the difference between its prediction and the model's average prediction. For tree ensembles, *TreeExplainer* computes exact SHAP values in polynomial time by following the tree structure. A feature's global importance is its mean absolute SHAP value over all samples: how much it moves the prediction on average, in either direction.

(c) Application. For each drug, the best model from Task 2 is refit on all of that drug's cell lines and explained with `shap.TreeExplainer`. The ten genes with the largest mean |SHAP| value are reported and shown as bar plots.

2.10 Software Tools

The analysis was written in Python. Table 1 lists the libraries used and their roles.

Table 1. Software tools and libraries used.

Tool	Role in this assignment
Python 3	Programming language for the whole pipeline
pandas	Loading the 331 MB CSV file, selecting each drug's rows, building result tables
NumPy	Numerical operations (averaging SHAP values, bar positions in plots)
scikit-learn	Decision Tree, Random Forest and GBM models; KFold; MAE, MSE, RMSE and R^2
XGBoost, LightGBM, CatBoost	The three specialised gradient boosting libraries
SHAP	TreeExplainer and the SHAP bar plots
Matplotlib	Bar plot of samples per drug and the per-drug metric charts

3. Data and Experimental Setup

3.1 Dataset

The dataset *hw4-drug-screening-data.csv* contains 12,502 rows and 2,532 columns. Each row is one cell line-drug pair. Table 2 summarises its columns.

Table 2. Columns of the drug screening dataset.

Column(s)	Meaning
CELL_LINE_NAME	Identifier of the cancer cell line (963 distinct cell lines)
DRUG_NAME	Name of the drug applied (13 distinct drugs)
LN_IC50	**Target.** Natural log of the IC50, the drug concentration needed to inhibit 50% of cell growth. Lower values mean the cell line is more sensitive.
2,529 gene columns	Gene expression levels of the cell line (A2M, AATF, ...), used as features

3.2 Pre-processing

- **Missing values:** every gene column was checked, and no gene is entirely missing.
- **Constant features:** genes with the same value in every row carry no information and were removed. One gene (KREMEN1) was dropped, leaving **2,528 genes** as features.
- **Scaling:** none was applied. Tree-based models split on thresholds and are not affected by the scale of the features.
- **Drug-specific data:** for each drug, only its rows (961-963 cell lines) were used. The drug name is not a feature, because it is constant within one drug's data. With 5 folds, each model is trained on about 770 cell lines and tested on about 192.

3.3 Choice of Metric for Selecting the Best Model

All four metrics are reported, but the best model per drug is selected by the lowest cross-validated RMSE, for these reasons:

- **R^2 would select the same models.** On a given test set, $R^2 = 1 - MSE / Var(y_{test})$. All models of a drug are tested on the same folds, so $Var(y_{test})$ is the same for all of them: a lower MSE means a

higher R^2, and RMSE is just the square root of MSE. In these results, MSE, RMSE and R^2 select the same best model for all 13 drugs.

- **Combining the metrics adds little.** MSE, RMSE and R^2 carry the same information. MAE is the only one that differs. It selects a different model for 3 drugs (5-Fluorouracil, Pictilisib, Staurosporine), but by margins of about 0.005, well within the variation between folds.
- **RMSE suits the task.** It is in the units of the target (log IC50), it penalises large errors, which matter most when predicting drug sensitivity, and it is the usual metric for this type of data.
- **R^2 is used to compare drugs.** Because it is unitless, R^2 can be compared across drugs whose LN_IC50 values have different spreads, while RMSE cannot.

4. Results

4.1 Task 1: Distribution of Samples per Drug

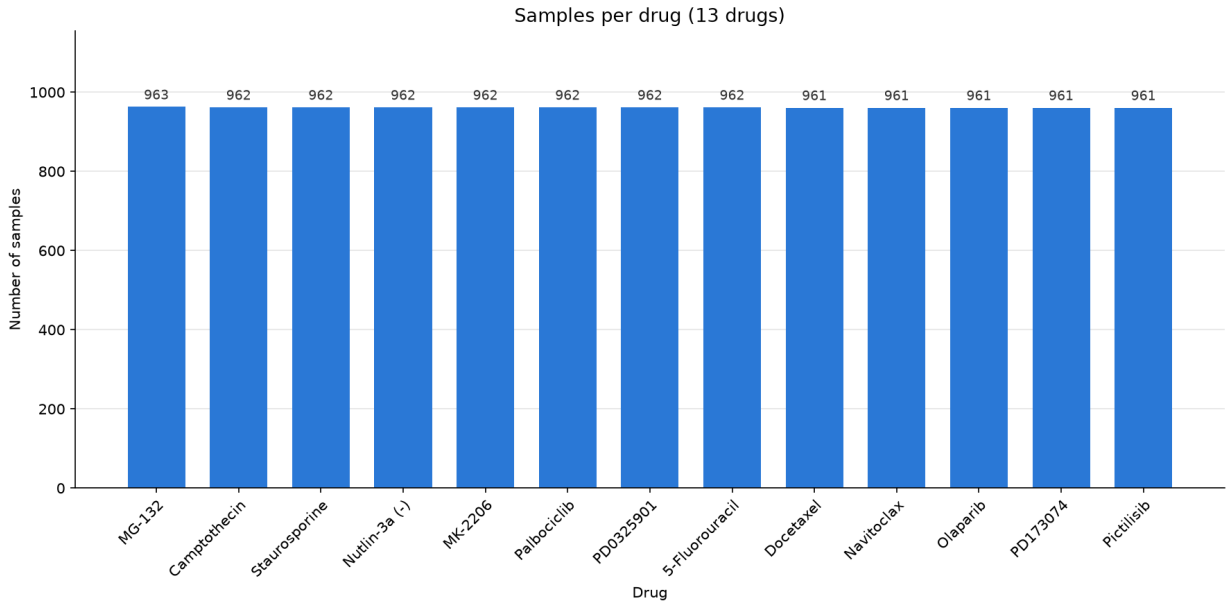


Figure 1. Number of samples (cell lines) for each of the 13 drugs.

Description

Figure 1 is a bar plot with one bar per drug, sorted by sample count. The height of each bar and the label above it give the number of cell lines screened with that drug.

Observations

- The dataset is almost perfectly balanced: every drug has between 961 and 963 samples. MG-132 has the most (963). Seven drugs have 962, and Docetaxel, Navitoclax, Olaparib, PD173074 and Pictilisib have 961.
- In total, 13 drugs x about 962 cell lines give the 12,502 rows of the dataset.
- Because the counts are nearly identical, every drug-specific model is trained on about the same amount of data. Differences in model performance between drugs therefore reflect how predictable each drug is, not differences in sample size.

Conclusion

The data contains enough samples (about 960) for each drug to train and cross-validate a separate model per drug, and no re-balancing is needed.

4.2 Task 2: Drug-Specific Regression Analysis

Tables 3-15 give the 5-fold cross-validated metrics of the six regressors for each drug, and Figures 2-14 show them as grouped bar charts. Bold values and boxed labels mark the best value of each metric; the shaded row is the best model (lowest RMSE). Lower is better for MAE, MSE and RMSE; higher is better for R^2.

Drug: 5-Fluorouracil

Table 3. Cross-validated performance of the six regressors for 5-Fluorouracil (best model: CatBoost).

Model	MAE	MSE	RMSE	R^2
Decision Tree	1.399	3.070	1.750	-0.062
Random Forest	1.154	2.073	1.436	0.290
Gradient Boosting Machine	**1.133**	2.020	1.418	0.305
XGBoost	1.143	2.042	1.425	0.300
LightGBM	1.163	2.084	1.440	0.284
CatBoost	1.137	**2.016**	**1.416**	**0.309**

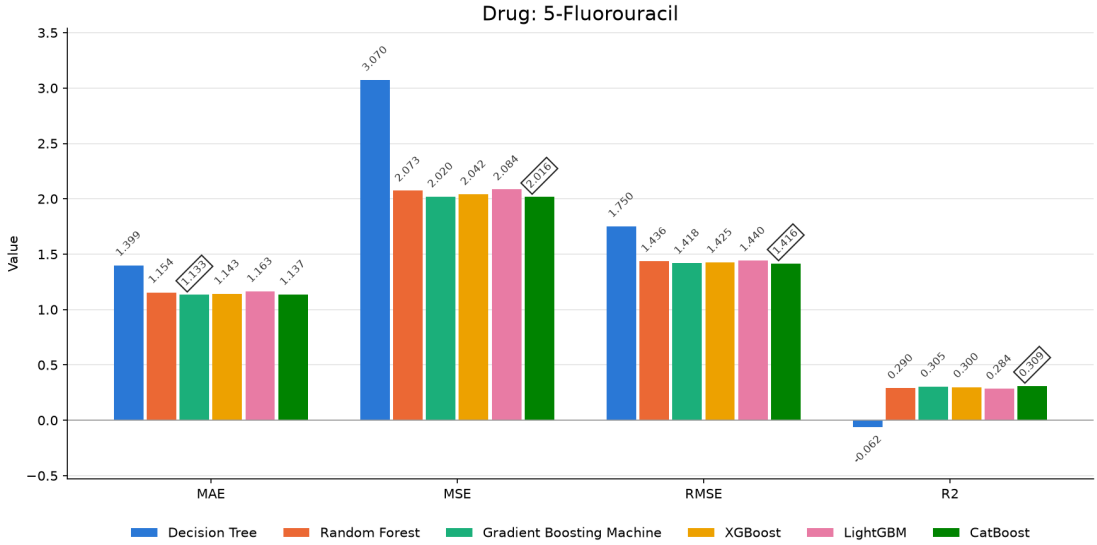


Figure 2. MAE, MSE, RMSE and R^2 of the six regressors for 5-Fluorouracil.

Description. Table 3 and Figure 2 compare the six regressors on the 962 cell lines screened with 5-Fluorouracil. The best model is **CatBoost**, with RMSE 1.416, MAE 1.137 and R^2 0.309.

Observations.

- CatBoost has the lowest RMSE (1.416) and MSE and the highest R^2 (0.309), so it is selected as the best regressor.
- The Gradient Boosting Machine is practically tied with it: its RMSE is only 0.002 higher and it actually has the lowest MAE (1.133 vs 1.137). The five ensemble models all lie within 0.025 RMSE of each other.
- The single Decision Tree has a negative R^2 (-0.062): on held-out cell lines it predicts worse than simply using the mean LN_IC50.

Drug: Camptothecin

Table 4. Cross-validated performance of the six regressors for Camptothecin (best model: XGBoost).

Model	MAE	MSE	RMSE	R^2
Decision Tree	1.365	2.959	1.718	0.114
Random Forest	1.214	2.306	1.518	0.314
Gradient Boosting Machine	1.178	2.237	1.495	0.333
XGBoost	**1.162**	**2.204**	**1.484**	**0.344**
LightGBM	1.176	2.255	1.501	0.329
CatBoost	1.176	2.224	1.490	0.337

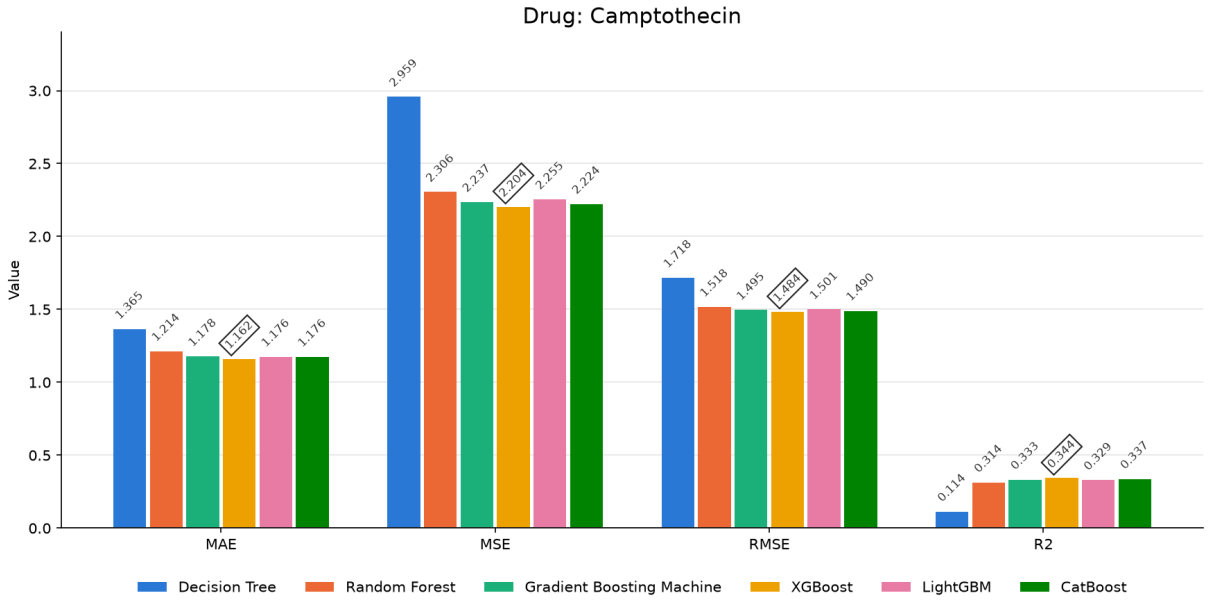


Figure 3. MAE, MSE, RMSE and R^2 of the six regressors for Camptothecin.

Description. Table 4 and Figure 3 compare the six regressors on the 962 cell lines screened with Camptothecin. The best model is **XGBoost**, with RMSE 1.484, MAE 1.162 and R^2 0.344.

Observations.

- XGBoost is the best model on all four metrics (RMSE 1.484, R^2 0.344).
- CatBoost, the Gradient Boosting Machine and LightGBM follow within 0.017 RMSE, and Random Forest is slightly behind (R^2 0.314).
- The Decision Tree is still clearly the weakest model, but it keeps a small positive R^2 (0.114).

Drug: Docetaxel

Table 5. Cross-validated performance of the six regressors for Docetaxel (best model: CatBoost).

Model	MAE	MSE	RMSE	R^2
Decision Tree	1.259	2.725	1.650	-0.114
Random Forest	1.070	1.859	1.362	0.241
Gradient Boosting Machine	1.015	1.712	1.307	0.302
XGBoost	1.020	1.715	1.308	0.301
LightGBM	1.007	1.693	1.299	0.311
CatBoost	**1.000**	**1.674**	**1.292**	**0.318**

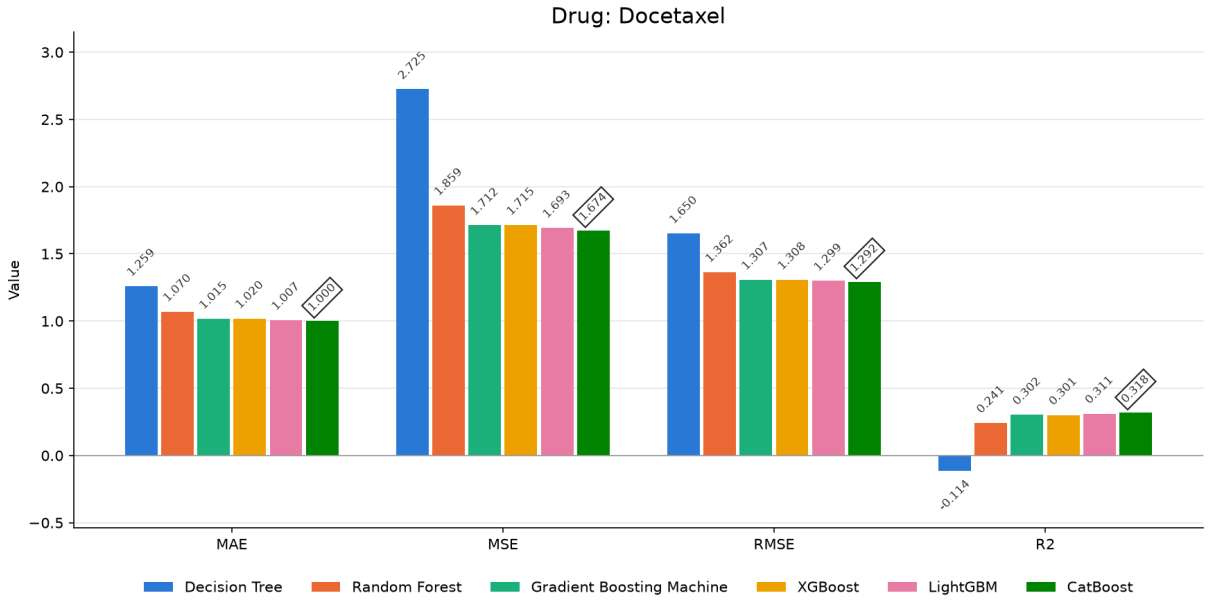


Figure 4. MAE, MSE, RMSE and R^2 of the six regressors for Docetaxel.

Description. Table 5 and Figure 4 compare the six regressors on the 961 cell lines screened with Docetaxel. The best model is **CatBoost**, with RMSE 1.292, MAE 1.000 and R^2 0.318.

Observations.

- CatBoost is the best model on all four metrics (RMSE 1.292, R^2 0.318), with LightGBM a close second (RMSE 1.299).
- The gap between Random Forest (R^2 0.241) and the boosting models (0.30-0.32) is wider than for most drugs, so building trees sequentially helps noticeably here.
- The Decision Tree has a negative R^2 (-0.114).

Drug: MG-132

Table 6. Cross-validated performance of the six regressors for MG-132 (best model: CatBoost).

Model	MAE	MSE	RMSE	R^2
Decision Tree	0.519	0.538	0.733	-0.052
Random Forest	0.456	0.429	0.652	0.175
Gradient Boosting Machine	0.448	0.420	0.646	0.186
XGBoost	0.448	0.413	0.641	0.198
LightGBM	0.452	0.423	0.648	0.178
CatBoost	**0.446**	**0.408**	**0.637**	**0.211**

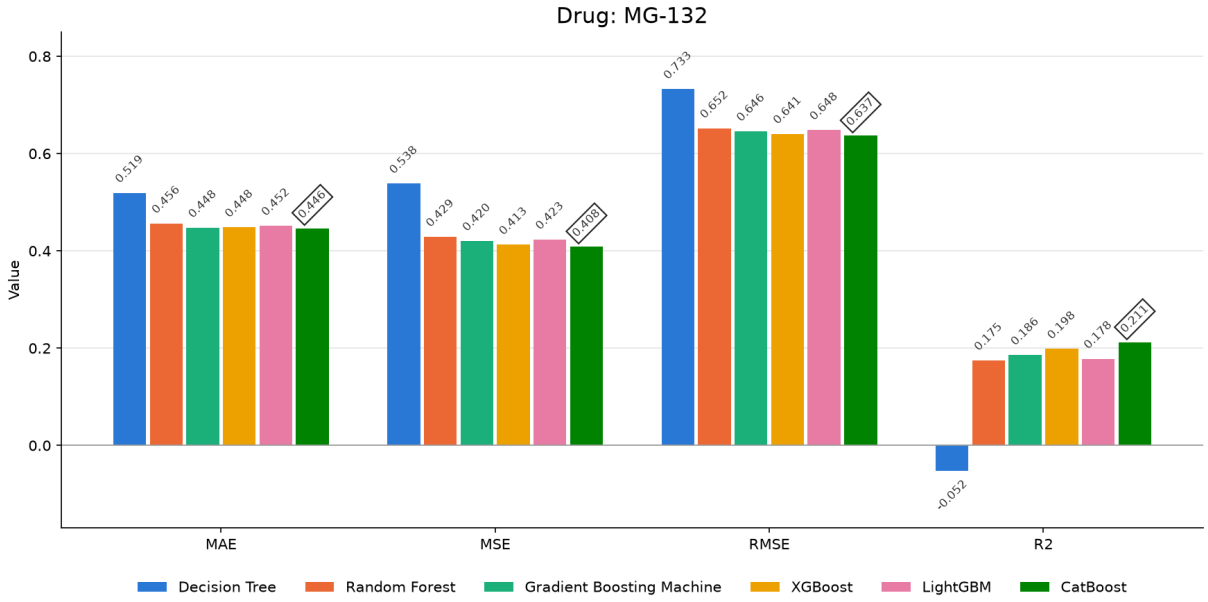


Figure 5. MAE, MSE, RMSE and R^2 of the six regressors for MG-132.

Description. Table 6 and Figure 5 compare the six regressors on the 963 cell lines screened with MG-132. The best model is **CatBoost**, with RMSE 0.637, MAE 0.446 and R^2 0.211.

Observations.

- CatBoost is the best model on all four metrics (RMSE 0.637, R^2 0.211).
- MG-132 has the smallest errors of all 13 drugs (MAE about 0.45), yet one of the lowest R^2 values. Its LN_IC50 values have a narrow spread, so small absolute errors are still a large share of the variance. This is why R^2, not RMSE, is used to compare predictability across drugs.
- All five ensembles are close (RMSE 0.637-0.652); the Decision Tree has a negative R^2 (-0.052).

Drug: MK-2206

Table 7. Cross-validated performance of the six regressors for MK-2206 (best model: LightGBM).

Model	MAE	MSE	RMSE	R^2
Decision Tree	1.199	2.327	1.523	-0.071
Random Forest	1.002	1.618	1.271	0.255
Gradient Boosting Machine	1.002	1.612	1.268	0.254
XGBoost	0.997	1.579	1.256	0.269
LightGBM	**0.991**	**1.579**	**1.256**	**0.271**
CatBoost	1.003	1.596	1.262	0.263

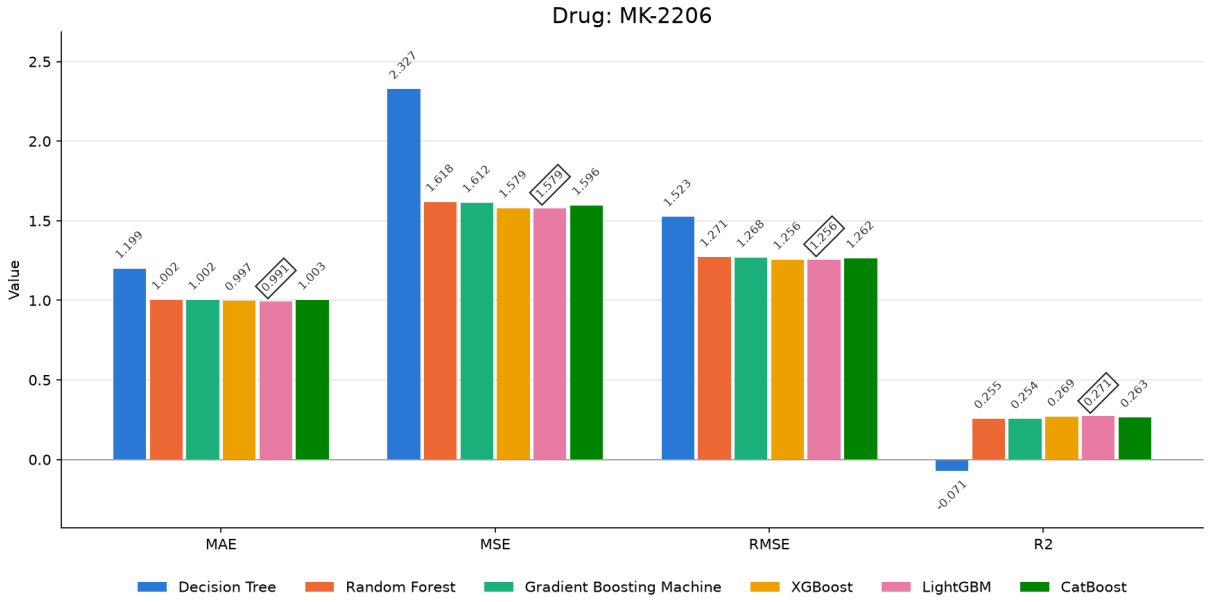


Figure 6. MAE, MSE, RMSE and R^2 of the six regressors for MK-2206.

Description. Table 7 and Figure 6 compare the six regressors on the 962 cell lines screened with MK-2206. The best model is **LightGBM**, with RMSE 1.256, MAE 0.991 and R^2 0.271.

Observations.

- LightGBM has the lowest RMSE and MAE and the highest R^2 (0.271), but XGBoost is essentially tied (RMSE 1.2556 vs 1.2557).
- Random Forest, CatBoost and the Gradient Boosting Machine are all within 0.015 RMSE of the best model, so the choice of ensemble matters little for this drug.
- The Decision Tree has a negative R^2 (-0.071).

Drug: Navitoclax

Table 8. Cross-validated performance of the six regressors for Navitoclax (best model: XGBoost).

Model	MAE	MSE	RMSE	R^2
Decision Tree	1.321	2.962	1.719	0.254
Random Forest	1.208	2.366	1.538	0.405
Gradient Boosting Machine	1.121	2.068	1.435	0.481
XGBoost	**1.109**	**2.050**	**1.429**	**0.486**
LightGBM	1.129	2.101	1.447	0.472
CatBoost	1.122	2.068	1.436	0.481

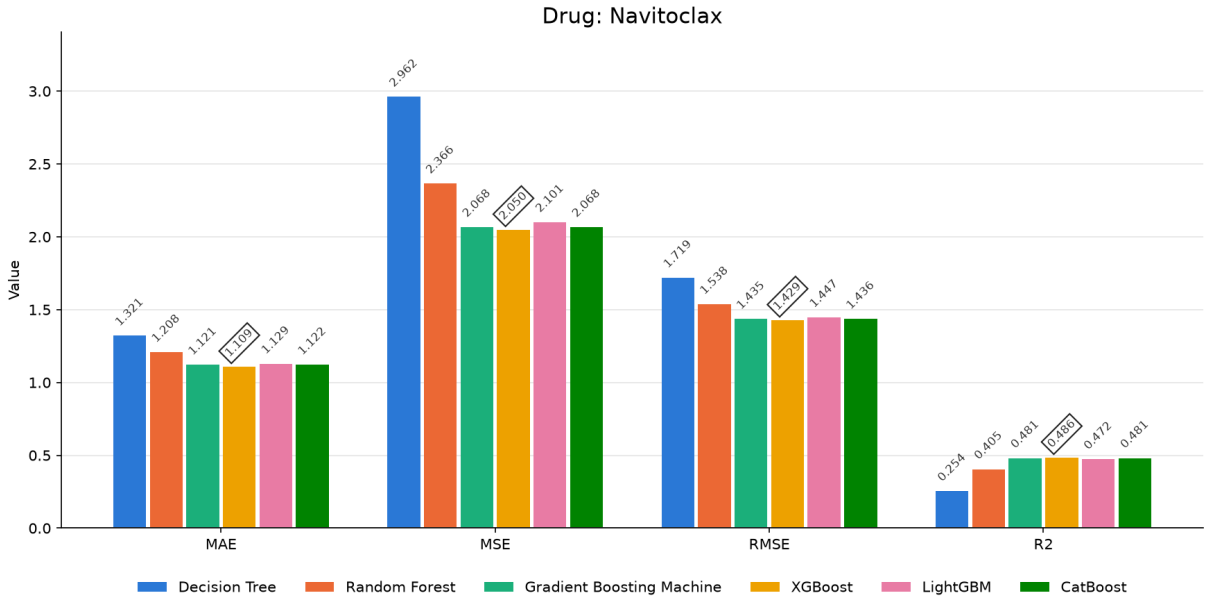


Figure 7. MAE, MSE, RMSE and R^2 of the six regressors for Navitoclax.

Description. Table 8 and Figure 7 compare the six regressors on the 961 cell lines screened with Navitoclax. The best model is **XGBoost**, with RMSE 1.429, MAE 1.109 and R^2 0.486.

Observations.

- XGBoost is the best model on all four metrics (RMSE 1.429, R^2 0.486).
- The Gradient Boosting Machine and CatBoost are almost identical (R^2 0.481), and Random Forest is clearly behind (0.405).
- The Decision Tree reaches R^2 0.254, one of its best results. This suggests a few strong genes (BCL2 and MCL1, see Section 4.3) that even a single tree can split on.

Drug: Nutlin-3a (-)

Table 9. Cross-validated performance of the six regressors for Nutlin-3a (-) (best model: LightGBM).

Model	MAE	MSE	RMSE	R^2
Decision Tree	1.160	2.437	1.555	0.233
Random Forest	1.014	1.708	1.305	0.457
Gradient Boosting Machine	0.867	1.297	1.137	0.585
XGBoost	0.862	1.294	1.137	0.586
LightGBM	**0.860**	**1.264**	**1.123**	**0.596**
CatBoost	0.873	1.292	1.136	0.586

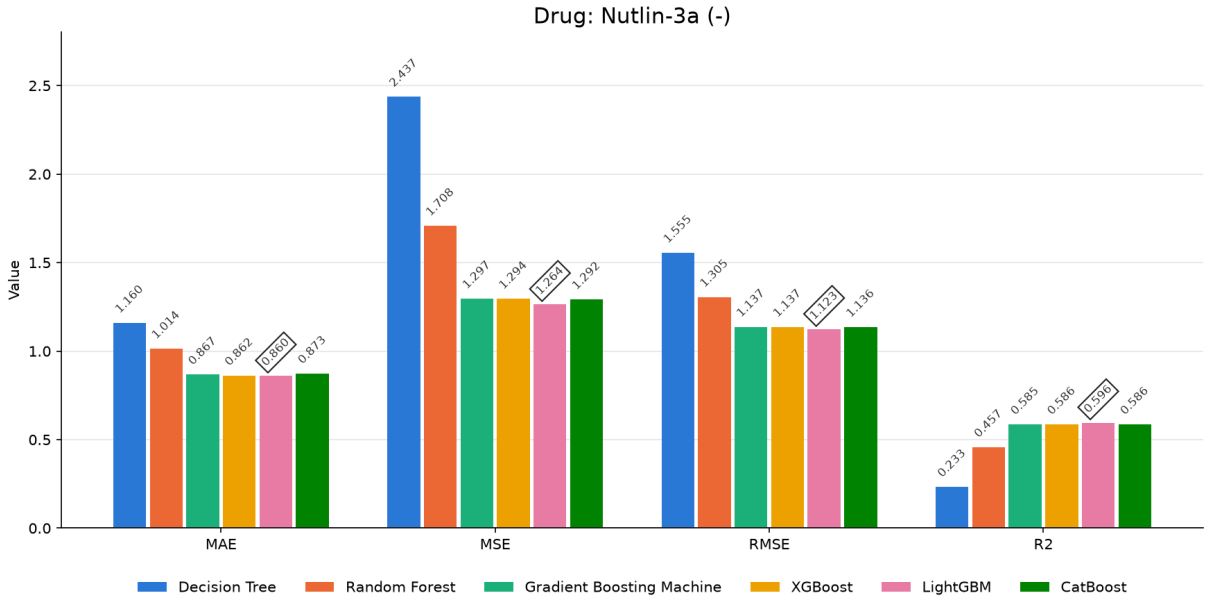


Figure 8. MAE, MSE, RMSE and R^2 of the six regressors for Nutlin-3a (-).

Description. Table 9 and Figure 8 compare the six regressors on the 962 cell lines screened with Nutlin-3a (-). The best model is **LightGBM**, with RMSE 1.123, MAE 0.860 and R^2 0.596.

Observations.

- LightGBM is the best model (RMSE 1.123) and gives the highest R^2 of the whole study (0.596): gene expression explains about 60% of the variation in response to Nutlin-3a.
- The four boosting models lie within 0.014 RMSE of each other, while Random Forest is far behind (R^2 0.457) and the Decision Tree reaches 0.233.
- This is the drug where boosting has the largest advantage over Random Forest.

Drug: Olaparib

Table 10. Cross-validated performance of the six regressors for Olaparib (best model: Gradient Boosting Machine).

Model	MAE	MSE	RMSE	R^2
Decision Tree	0.916	1.363	1.165	0.179
Random Forest	0.749	0.948	0.972	0.432
Gradient Boosting Machine	**0.724**	**0.873**	**0.933**	**0.477**
XGBoost	0.726	0.878	0.935	0.474
LightGBM	0.730	0.881	0.937	0.472
CatBoost	0.730	0.884	0.939	0.470

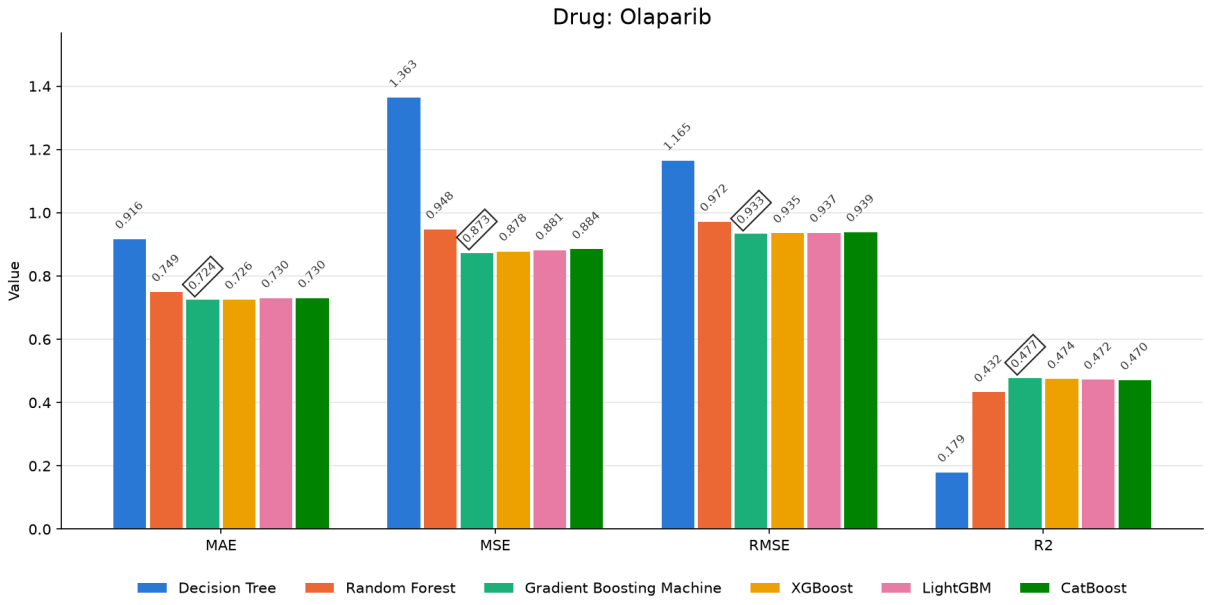


Figure 9. MAE, MSE, RMSE and R^2 of the six regressors for Olaparib.

Description. Table 10 and Figure 9 compare the six regressors on the 961 cell lines screened with Olaparib. The best model is **Gradient Boosting Machine**, with RMSE 0.933, MAE 0.724 and R^2 0.477.

Observations.

- The Gradient Boosting Machine is the best model on all four metrics (RMSE 0.933, R^2 0.477).
- The four boosting models are within 0.006 RMSE, which is effectively a tie. Random Forest follows with R^2 0.432.
- The Decision Tree keeps a positive R^2 (0.179) but is clearly the weakest model.

Drug: PD0325901

Table 11. Cross-validated performance of the six regressors for PD0325901 (best model: LightGBM).

Model	MAE	MSE	RMSE	R^2
Decision Tree	1.504	4.011	1.997	0.125
Random Forest	1.253	2.705	1.636	0.415
Gradient Boosting Machine	1.164	2.440	1.554	0.471
XGBoost	1.154	2.390	1.539	0.481
LightGBM	**1.148**	**2.354**	**1.526**	**0.491**
CatBoost	1.150	2.409	1.545	0.477

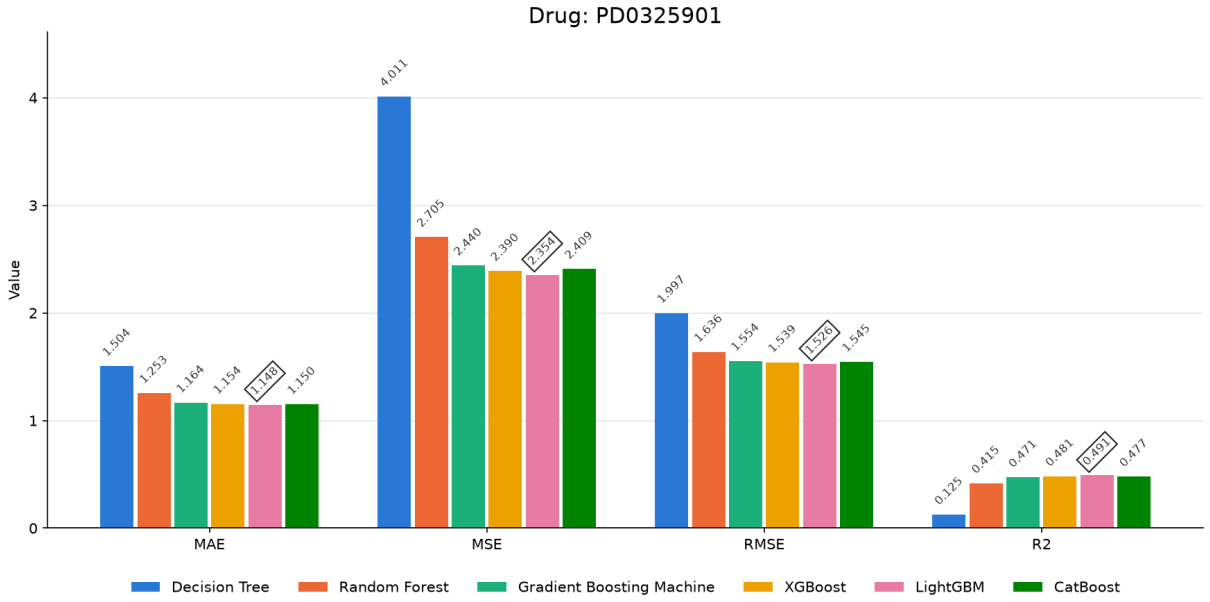


Figure 10. MAE, MSE, RMSE and R^2 of the six regressors for PD0325901.

Description. Table 11 and Figure 10 compare the six regressors on the 962 cell lines screened with PD0325901. The best model is **LightGBM**, with RMSE 1.526, MAE 1.148 and R^2 0.491.

Observations.

- LightGBM is the best model on all four metrics (RMSE 1.526, R^2 0.491).
- Its RMSE is the highest among the 13 best models, but its R^2 is the second highest. LN_IC50 varies widely for this MEK inhibitor, so the errors are large in absolute terms but small relative to the spread of the data.
- The Decision Tree has the largest error of any model in the study (RMSE 1.997).

Drug: PD173074

Table 12. Cross-validated performance of the six regressors for PD173074 (best model: CatBoost).

Model	MAE	MSE	RMSE	R^2
Decision Tree	1.123	2.144	1.462	-0.109
Random Forest	0.943	1.597	1.259	0.185
Gradient Boosting Machine	0.918	1.412	1.185	0.275
XGBoost	0.932	1.463	1.205	0.251
LightGBM	0.958	1.584	1.254	0.189
CatBoost	**0.913**	**1.404**	**1.181**	**0.280**

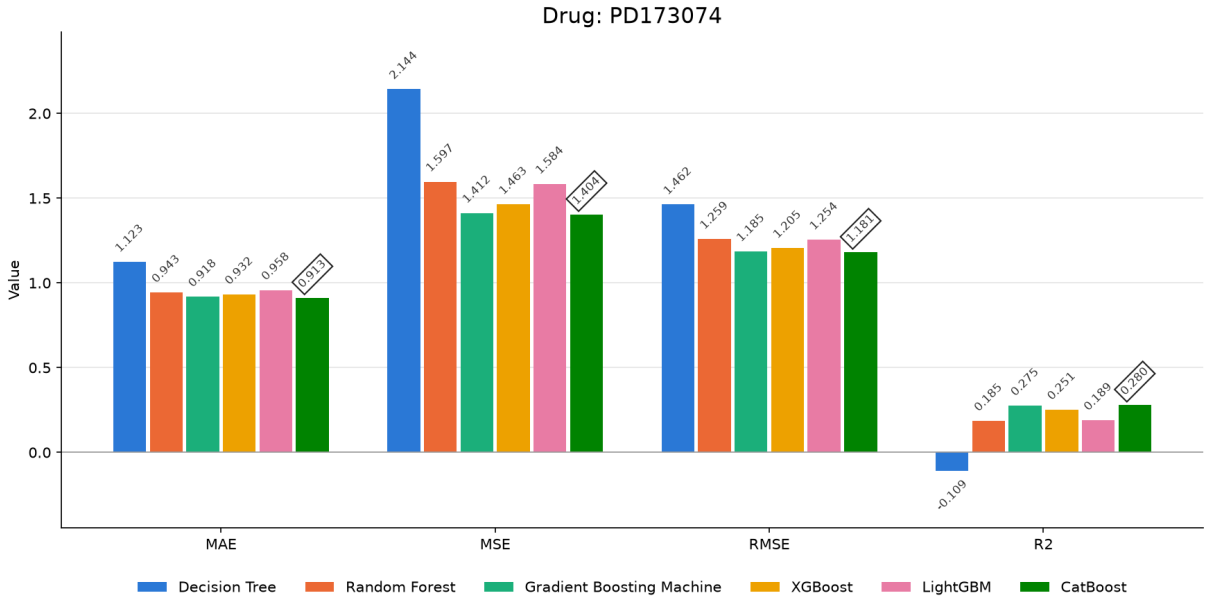


Figure 11. MAE, MSE, RMSE and R^2 of the six regressors for PD173074.

Description. Table 12 and Figure 11 compare the six regressors on the 961 cell lines screened with PD173074. The best model is **CatBoost**, with RMSE 1.181, MAE 0.913 and R^2 0.280.

Observations.

- CatBoost is the best model on all four metrics (RMSE 1.181, R^2 0.280), closely followed by the Gradient Boosting Machine (0.275).
- The models are more spread out than for other drugs: LightGBM (R^2 0.189) performs about as poorly as Random Forest (0.185).
- The Decision Tree has a negative R^2 (-0.109).

Drug: Palbociclib

Table 13. Cross-validated performance of the six regressors for Palbociclib (best model: CatBoost).

Model	MAE	MSE	RMSE	R^2
Decision Tree	1.145	2.098	1.448	0.141
Random Forest	0.937	1.413	1.188	0.424
Gradient Boosting Machine	0.917	1.362	1.167	0.444
XGBoost	0.910	1.348	1.161	0.450
LightGBM	0.911	1.346	1.160	0.450
CatBoost	**0.909**	**1.321**	**1.149**	**0.461**

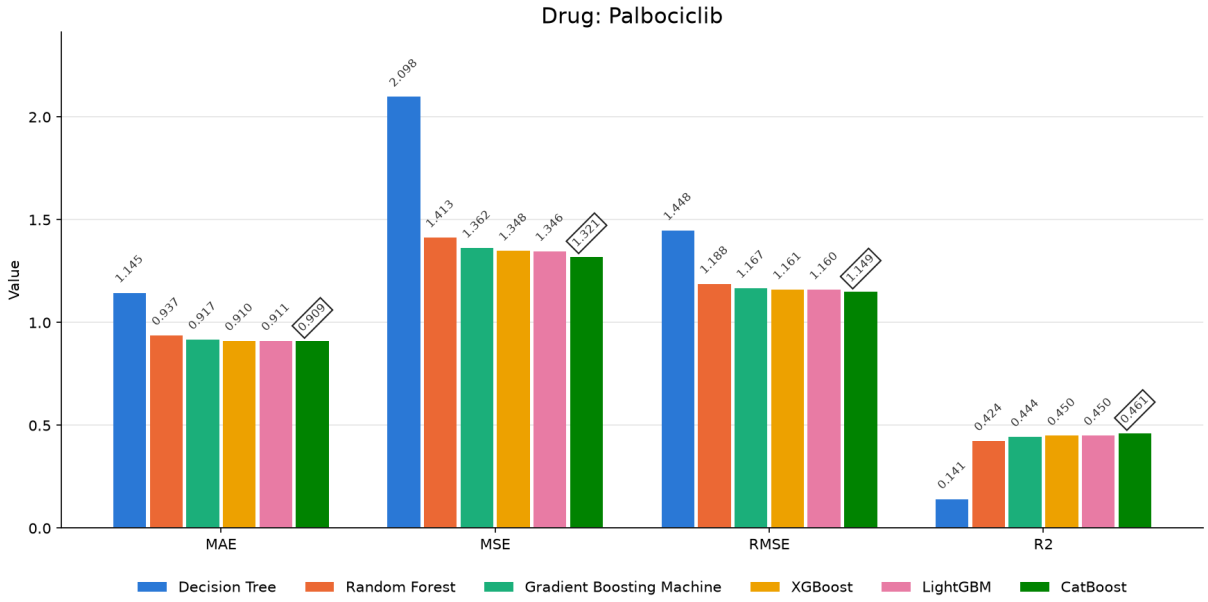


Figure 12. MAE, MSE, RMSE and R^2 of the six regressors for Palbociclib.

Description. Table 13 and Figure 12 compare the six regressors on the 962 cell lines screened with Palbociclib. The best model is **CatBoost**, with RMSE 1.149, MAE 0.909 and R^2 0.461.

Observations.

- CatBoost is the best model on all four metrics (RMSE 1.149, R^2 0.461).
- LightGBM and XGBoost tie for second place (R^2 0.450). Random Forest is closer to the boosting models here than for most drugs (0.424).
- The Decision Tree reaches a small positive R^2 (0.141).

Drug: Pictilisib

Table 14. Cross-validated performance of the six regressors for Pictilisib (best model: XGBoost).

Model	MAE	MSE	RMSE	R^2
Decision Tree	1.043	1.789	1.338	-0.144
Random Forest	0.912	1.378	1.172	0.126
Gradient Boosting Machine	**0.903**	1.340	1.156	0.149
XGBoost	0.904	**1.335**	**1.154**	**0.152**
LightGBM	0.910	1.355	1.163	0.140
CatBoost	0.903	1.355	1.163	0.139

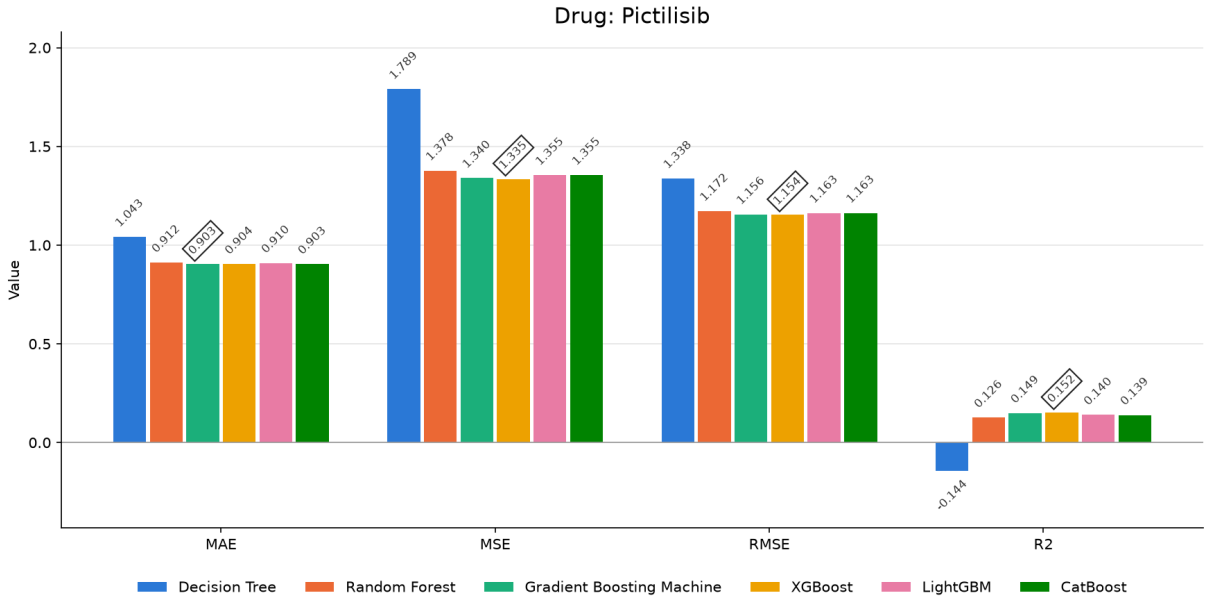


Figure 13. MAE, MSE, RMSE and R^2 of the six regressors for Pictilisib.

Description. Table 14 and Figure 13 compare the six regressors on the 961 cell lines screened with Pictilisib. The best model is **XGBoost**, with RMSE 1.154, MAE 0.904 and R^2 0.152.

Observations.

- XGBoost has the lowest RMSE (1.154) and the highest R^2 (0.152), while the Gradient Boosting Machine and CatBoost have a marginally lower MAE (0.903 vs 0.904).
- Pictilisib is the hardest drug to predict: the best model explains only about 15% of the variance, and all five ensembles fall between 0.126 and 0.152.
- The Decision Tree has the most negative R^2 of the study (-0.144).

Drug: Staurosporine

Table 15. Cross-validated performance of the six regressors for Staurosporine (best model: Gradient Boosting Machine).

Model	MAE	MSE	RMSE	R^2
Decision Tree	1.107	1.999	1.414	-0.063
Random Forest	0.943	1.481	1.215	0.223
Gradient Boosting Machine	0.914	**1.380**	**1.173**	**0.274**
XGBoost	0.923	1.424	1.191	0.251
LightGBM	**0.907**	1.394	1.179	0.266
CatBoost	0.917	1.427	1.193	0.250

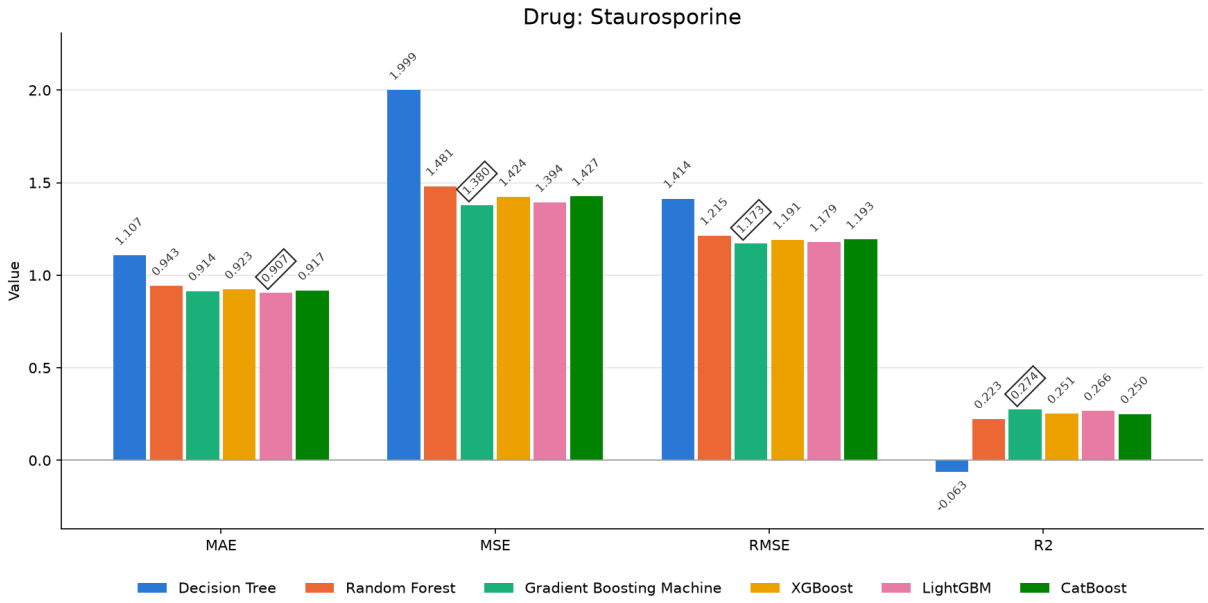


Figure 14. MAE, MSE, RMSE and R^2 of the six regressors for Staurosporine.

Description. Table 15 and Figure 14 compare the six regressors on the 962 cell lines screened with Staurosporine. The best model is **Gradient Boosting Machine**, with RMSE 1.173, MAE 0.914 and R^2 0.274.

Observations.

- The Gradient Boosting Machine has the lowest RMSE (1.173) and the highest R^2 (0.274). LightGBM has the lowest MAE (0.907 vs 0.914) and the second-best RMSE.
- XGBoost and CatBoost are about 0.02 RMSE behind, and Random Forest is further behind (R^2 0.223).
- The Decision Tree has a negative R^2 (-0.063).

Summary of Task 2

Table 16 collects the best regressor of each drug and its metrics.

Table 16. Best tree-based regressor for each drug (selected by lowest cross-validated RMSE).

Drug	Best model	MAE	MSE	RMSE	R^2
5-Fluorouracil	CatBoost	1.137	2.016	1.416	0.309
Camptothecin	XGBoost	1.162	2.204	1.484	0.344
Docetaxel	CatBoost	1.000	1.674	1.292	0.318
MG-132	CatBoost	0.446	0.408	0.637	0.211
MK-2206	LightGBM	0.991	1.579	1.256	0.271
Navitoclax	XGBoost	1.109	2.050	1.429	0.486
Nutlin-3a (-)	LightGBM	0.860	1.264	1.123	0.596
Olaparib	Gradient Boosting Machine	0.724	0.873	0.933	0.477
PD0325901	LightGBM	1.148	2.354	1.526	0.491
PD173074	CatBoost	0.913	1.404	1.181	0.280
Palbociclib	CatBoost	0.909	1.321	1.149	0.461
Pictilisib	XGBoost	0.904	1.335	1.154	0.152
Staurosporine	Gradient Boosting Machine	0.914	1.380	1.173	0.274

Description. Table 16 lists, for each of the 13 drugs, the regressor with the lowest RMSE and its four metrics.

Observations.

- A boosting model is the best regressor for every drug: CatBoost for 5 drugs, XGBoost for 3, LightGBM for 3 and the Gradient Boosting Machine for 2. Neither Random Forest nor the single Decision Tree wins for any drug.
- The four boosting models are usually within a few hundredths of RMSE of each other, so the "best" boosting library for a drug is often a near tie. The main gap is between boosting and the other models.
- R^2 of the best models ranges from 0.152 (Pictilisib) to 0.596 (Nutlin-3a). Five drugs (Nutlin-3a, PD0325901, Navitoclax, Olaparib and Palbociclib) reach R^2 of 0.46-0.60. The other eight stay between 0.15 and 0.34, with Pictilisib (0.152) and MG-132 (0.211) the hardest to predict.
- The single Decision Tree is the worst model for every drug and has a negative R^2 for 7 drugs. With about 770 training samples and 2,528 genes, one tree overfits to chance patterns. Averaging many trees (Random Forest) or building them gradually (boosting) generalises much better.
- MSE, RMSE and R^2 agree on the best model for all 13 drugs, and MAE agrees for 10 of them, so the conclusions do not depend on the choice of metric.

Conclusion. Gradient boosting ensembles are the most suitable tree-based regressors for predicting drug response from gene expression. Gene expression explains a substantial share (15-60%) of the variation in drug sensitivity, but not all of it.

4.3 Task 3: SHAP Analysis of the Best Regressors

For each drug, the best regressor from Table 16 was refit on all of that drug's cell lines and explained with SHAP TreeExplainer. Figures 15-27 show the ten genes with the largest mean absolute SHAP value for each drug. The x-axis is the mean |SHAP| value, which is the average amount, in LN_IC50 units, by which the gene moves the model's prediction away from the average prediction. Genes are sorted from most to least important.

Drug: 5-Fluorouracil

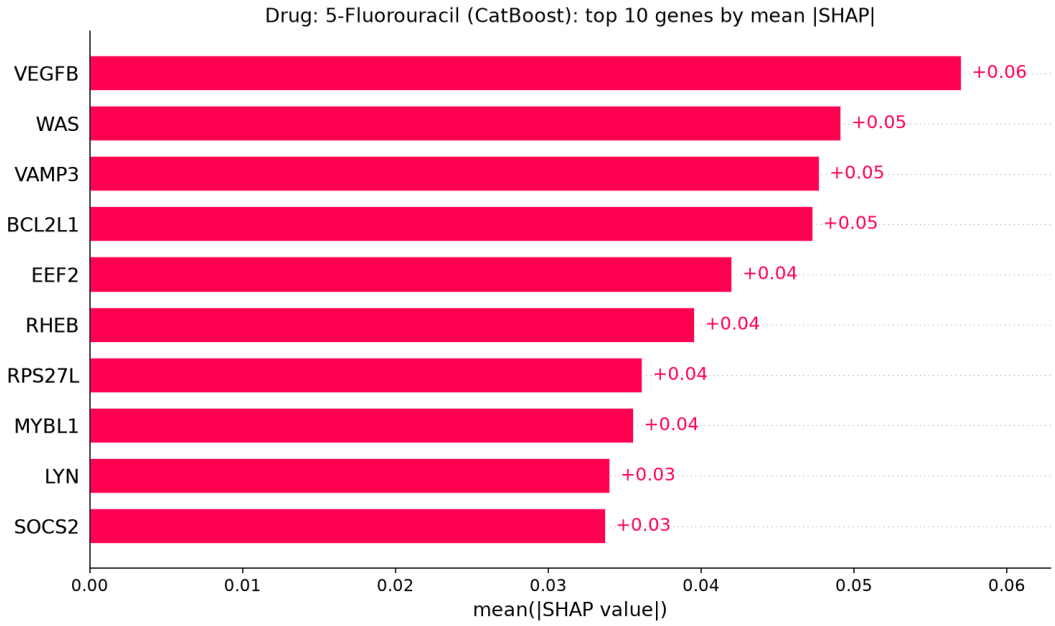


Figure 15. Top 10 genes by mean |SHAP| value for 5-Fluorouracil (best model: CatBoost).

Description. Figure 15 shows the SHAP importance of the ten most influential genes for the CatBoost model of 5-Fluorouracil. The three most important genes are VEGFB (0.057), WAS (0.049), VAMP3 (0.048); the tenth gene, SOCS2, has a mean |SHAP| of 0.034.

Observations.

- Importance is spread almost evenly over the ten genes (mean |SHAP| from 0.057 for VEGFB down to 0.034 for SOCS2), and no single gene dominates.
- This matches the drug's broad mechanism: 5-Fluorouracil is an antimetabolite that disrupts DNA and RNA synthesis in general rather than acting on one target. It also fits the moderate R^2 of 0.31.
- BCL2L1, an anti-apoptotic (cell-survival) gene, and RPS27L, a p53 target gene, both appear, which points to cell-death signalling as part of the response.

Drug: Camptothecin

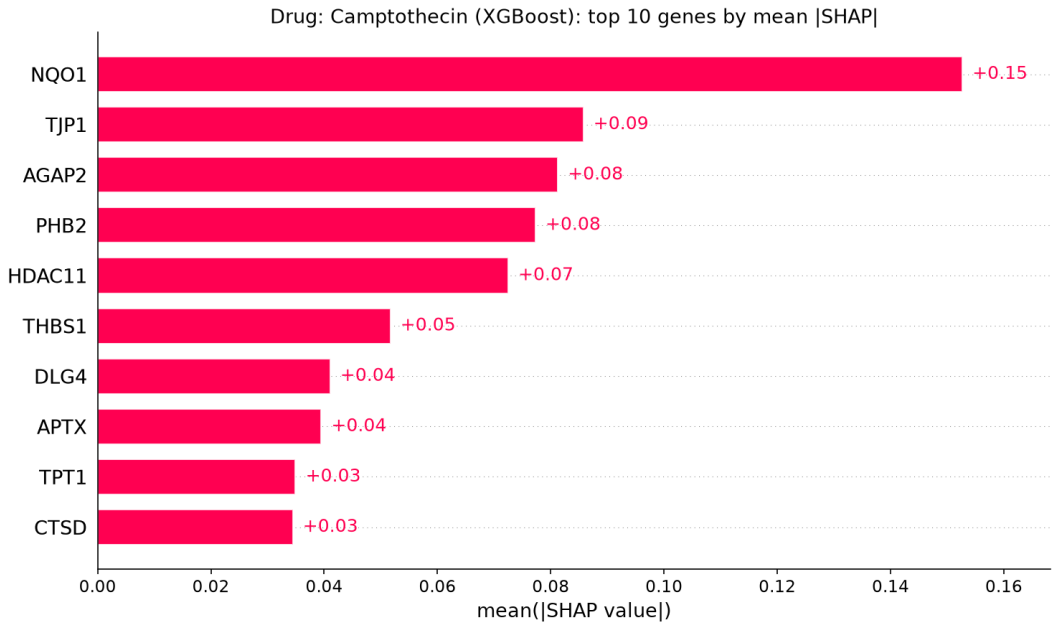


Figure 16. Top 10 genes by mean |SHAP| value for Camptothecin (best model: XGBoost).

Description. Figure 16 shows the SHAP importance of the ten most influential genes for the XGBoost model of Camptothecin. The three most important genes are NQO1 (0.153), TJP1 (0.086), AGAP2 (0.081); the tenth gene, CTSD, has a mean |SHAP| of 0.034.

Observations.

- NQO1 is by far the most important gene (0.153), almost twice the second gene, TJP1 (0.086).
- APTX (aprataxin), a DNA single-strand break repair enzyme, is ranked 8th. This is biologically plausible: camptothecin inhibits topoisomerase I and causes single-strand DNA breaks.
- After the top five genes, importance decreases gradually.

Drug: Docetaxel

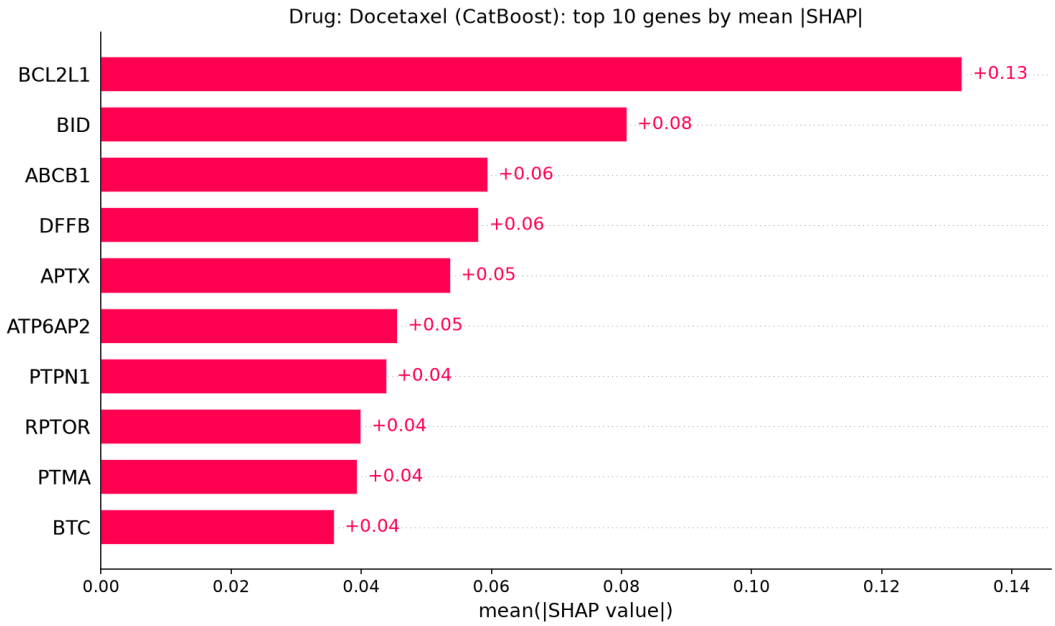


Figure 17. Top 10 genes by mean |SHAP| value for Docetaxel (best model: CatBoost).

Description. Figure 17 shows the SHAP importance of the ten most influential genes for the CatBoost model of Docetaxel. The three most important genes are BCL2L1 (0.132), BID (0.081), ABCB1 (0.060); the tenth gene, BTC, has a mean |SHAP| of 0.036.

Observations.

- BCL2L1 is the top gene (0.132), and the apoptosis genes BID and DFFB are ranked 2nd and 4th: the model links Docetaxel response to how easily the cells undergo programmed cell death.
- ABCB1 (3rd) encodes P-glycoprotein, the efflux pump that removes taxanes such as Docetaxel from cells and is a well-known cause of taxane resistance.

Drug: MG-132

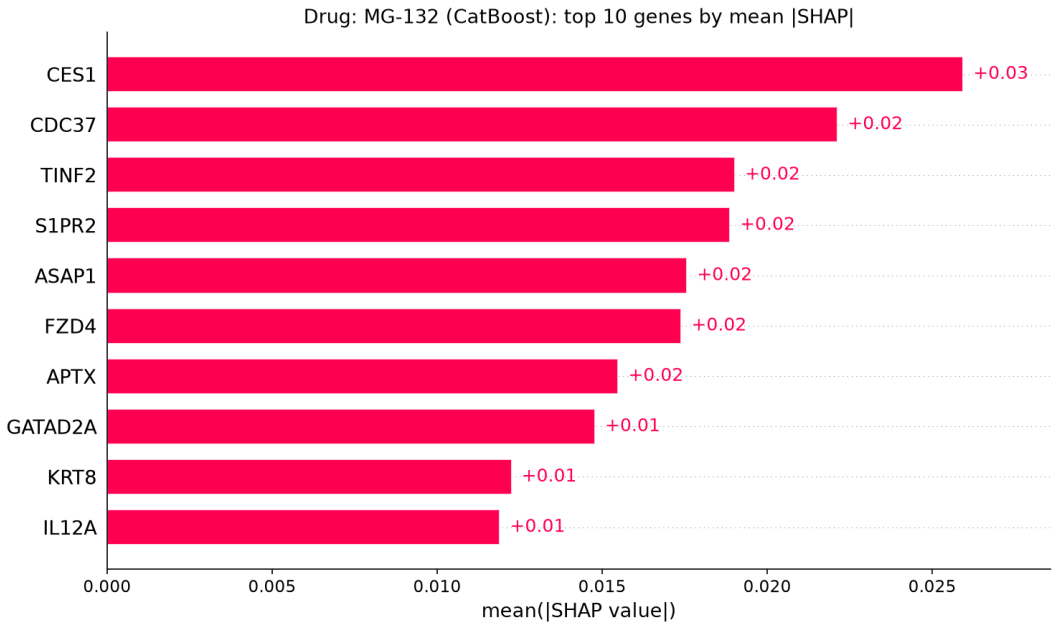


Figure 18. Top 10 genes by mean |SHAP| value for MG-132 (best model: CatBoost).

Description. Figure 18 shows the SHAP importance of the ten most influential genes for the CatBoost model of MG-132. The three most important genes are CES1 (0.026), CDC37 (0.022), TINF2 (0.019); the tenth gene, IL12A, has a mean |SHAP| of 0.012.

Observations.

- MG-132 has the smallest SHAP values of all drugs (the top gene, CES1, has only 0.026). This is consistent with the narrow LN_IC50 range and the low R^2 seen in Section 4.2.
- Importance is flat across the ten genes, so the model did not find a strong individual biomarker. CDC37 (2nd) is a co-chaperone of HSP90 involved in protein folding, which is plausibly related to proteasome inhibition.

Drug: MK-2206

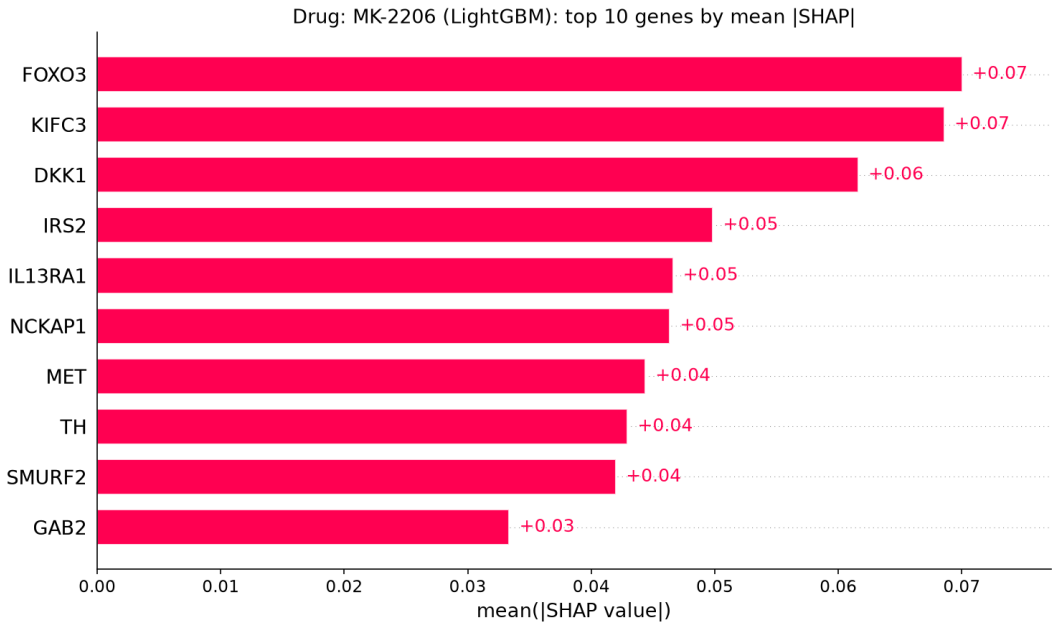


Figure 19. Top 10 genes by mean |SHAP| value for MK-2206 (best model: LightGBM).

Description. Figure 19 shows the SHAP importance of the ten most influential genes for the LightGBM model of MK-2206. The three most important genes are FOXO3 (0.070), KIFC3 (0.069), DKK1 (0.062); the tenth gene, GAB2, has a mean |SHAP| of 0.033.

Observations.

- MK-2206 is an AKT inhibitor, and its top gene FOXO3 is a direct substrate of AKT.
- IRS2 and GAB2 are adaptor proteins that activate PI3K/AKT signalling, and MET is a receptor tyrosine kinase upstream of the same pathway, so several top genes belong to the drug's target pathway.
- Importance is fairly flat (0.070 down to 0.033), matching the modest R^2 of 0.27.

Drug: Navitoclax

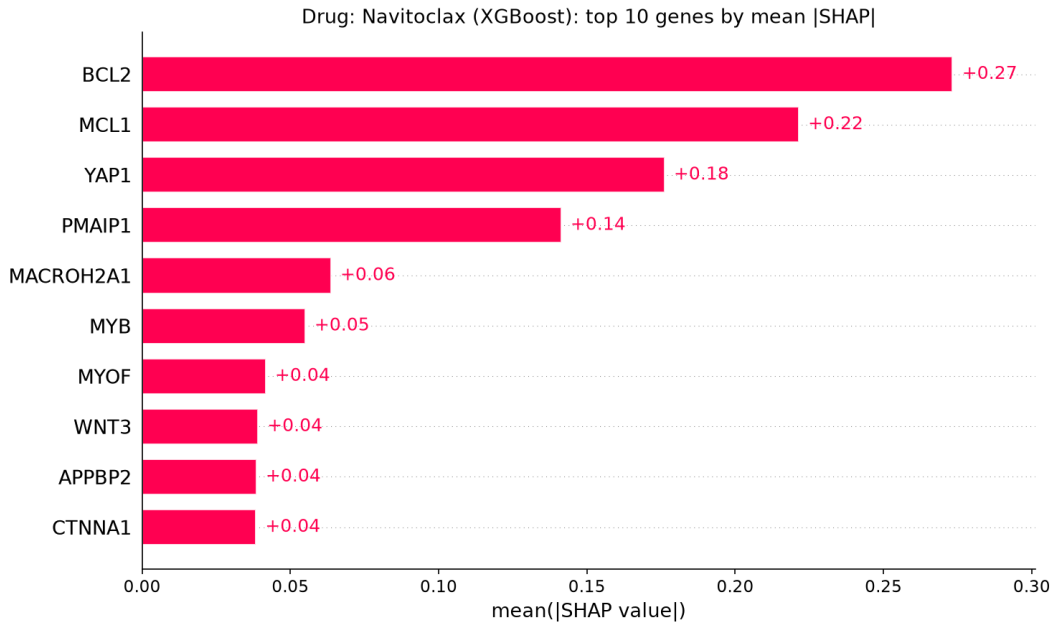


Figure 20. Top 10 genes by mean |SHAP| value for Navitoclax (best model: XGBoost).

Description. Figure 20 shows the SHAP importance of the ten most influential genes for the XGBoost model of Navitoclax. The three most important genes are BCL2 (0.273), MCL1 (0.221), YAP1 (0.176); the tenth gene, CTNNA1, has a mean |SHAP| of 0.038.

Observations.

- BCL2, the direct target of Navitoclax, is the most important gene, and it has the largest top-gene SHAP value of the study (0.273).
- MCL1 (2nd) is a related survival protein that Navitoclax does not inhibit and is a known cause of resistance. PMAIP1 (NOXA, 4th) inhibits MCL1.
- Importance drops sharply after the top four genes, showing that response is driven mainly by the BCL2-family balance.

Drug: Nutlin-3a (-)

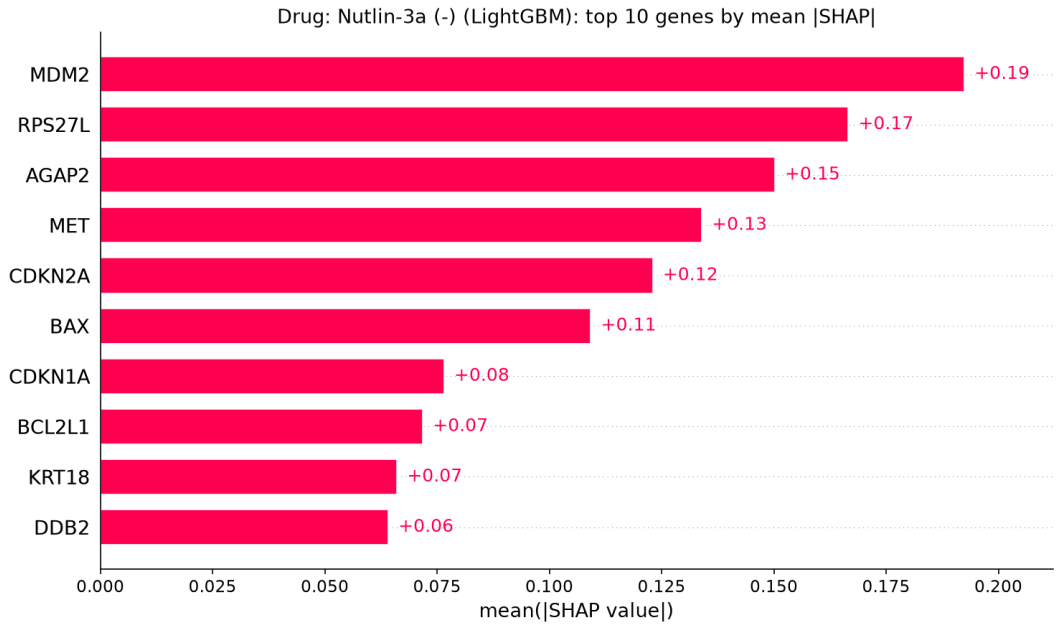


Figure 21. Top 10 genes by mean |SHAP| value for Nutlin-3a (-) (best model: LightGBM).

Description. Figure 21 shows the SHAP importance of the ten most influential genes for the LightGBM model of Nutlin-3a (-). The three most important genes are MDM2 (0.192), RPS27L (0.167), AGAP2 (0.150); the tenth gene, DDB2, has a mean |SHAP| of 0.064.

Observations.

- MDM2, the direct target of Nutlin-3a, is the most important gene (0.192).
- RPS27L, CDKN1A (p21), BAX and DDB2 are classic p53 target genes. Their expression indicates a functional (wild-type) p53, which Nutlin-3a needs in order to work, because it acts by releasing p53 from MDM2.
- All ten genes have relatively large SHAP values (at least 0.064), which agrees with this drug having the highest R^2 (0.60).

Drug: Olaparib

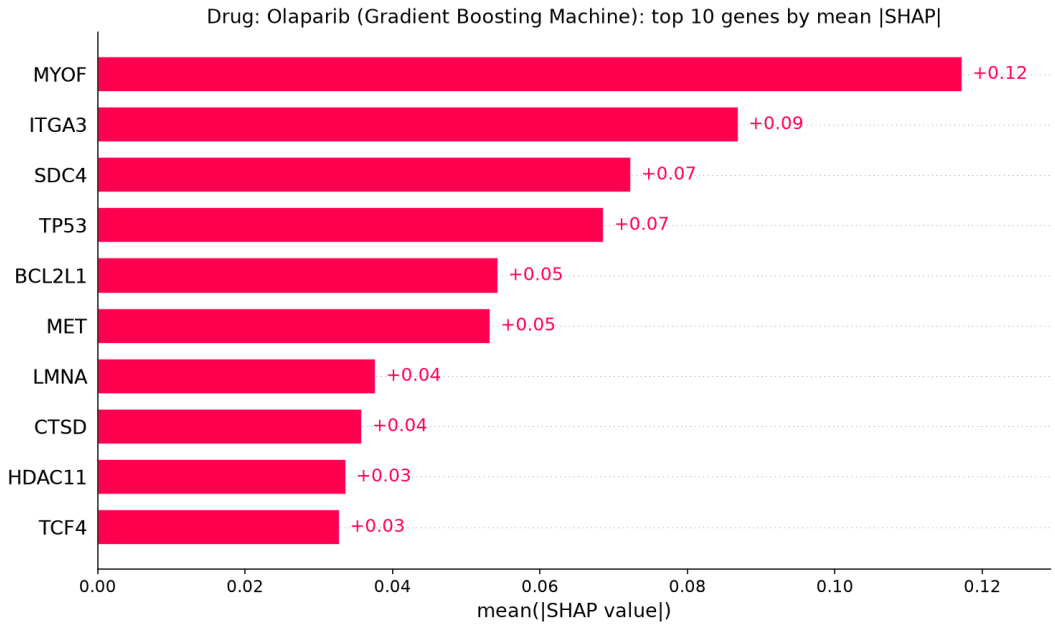


Figure 22. Top 10 genes by mean |SHAP| value for Olaparib (best model: Gradient Boosting Machine).

Description. Figure 22 shows the SHAP importance of the ten most influential genes for the Gradient Boosting Machine model of Olaparib. The three most important genes are MYOF (0.117), ITGA3 (0.087), SDC4 (0.072); the tenth gene, TCF4, has a mean |SHAP| of 0.033.

Observations.

- MYOF is the top gene (0.117), followed by ITGA3 and SDC4, which are cell-adhesion genes.
- TP53 (4th) is a central regulator of the DNA-damage response, which is relevant for a PARP inhibitor such as Olaparib.
- The mapping to the drug's mechanism is less direct than for the targeted drugs; some top genes may mark cell type rather than DNA-repair capacity.

Drug: PD0325901

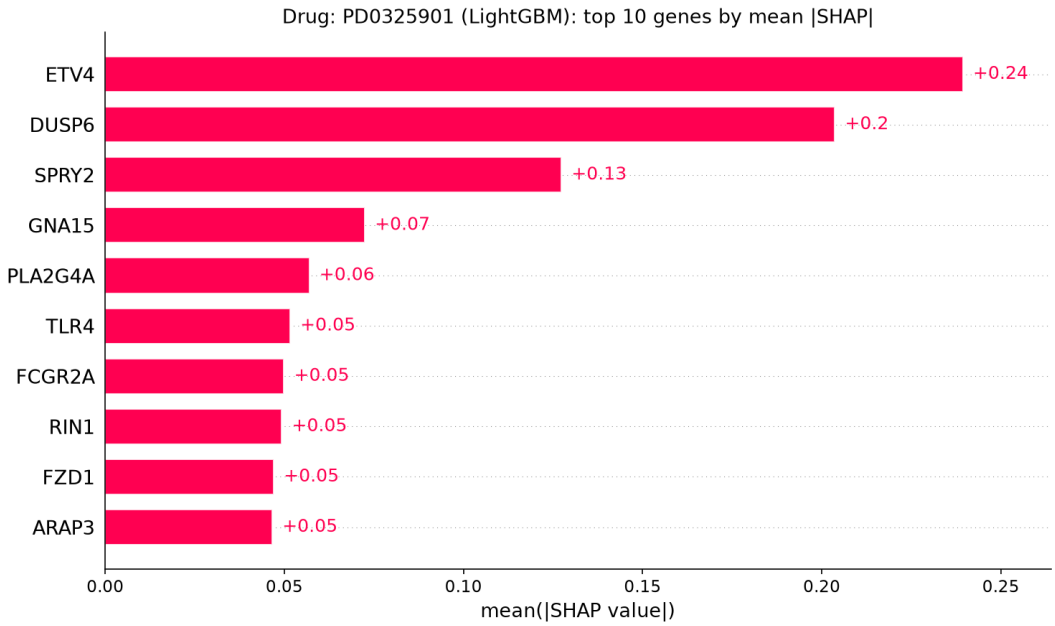


Figure 23. Top 10 genes by mean |SHAP| value for PD0325901 (best model: LightGBM).

Description. Figure 23 shows the SHAP importance of the ten most influential genes for the LightGBM model of PD0325901. The three most important genes are ETV4 (0.239), DUSP6 (0.204), SPRY2 (0.127); the tenth gene, ARAP3, has a mean |SHAP| of 0.047.

Observations.

- ETV4, DUSP6 and SPRY2 dominate the ranking (0.239, 0.204 and 0.127). These are standard transcriptional read-outs of MEK/ERK pathway activity.
- Cell lines with high expression of these genes depend on MAPK signalling and are therefore more sensitive to the MEK inhibitor PD0325901. This is a clear, mechanism-consistent biomarker set.

Drug: PD173074

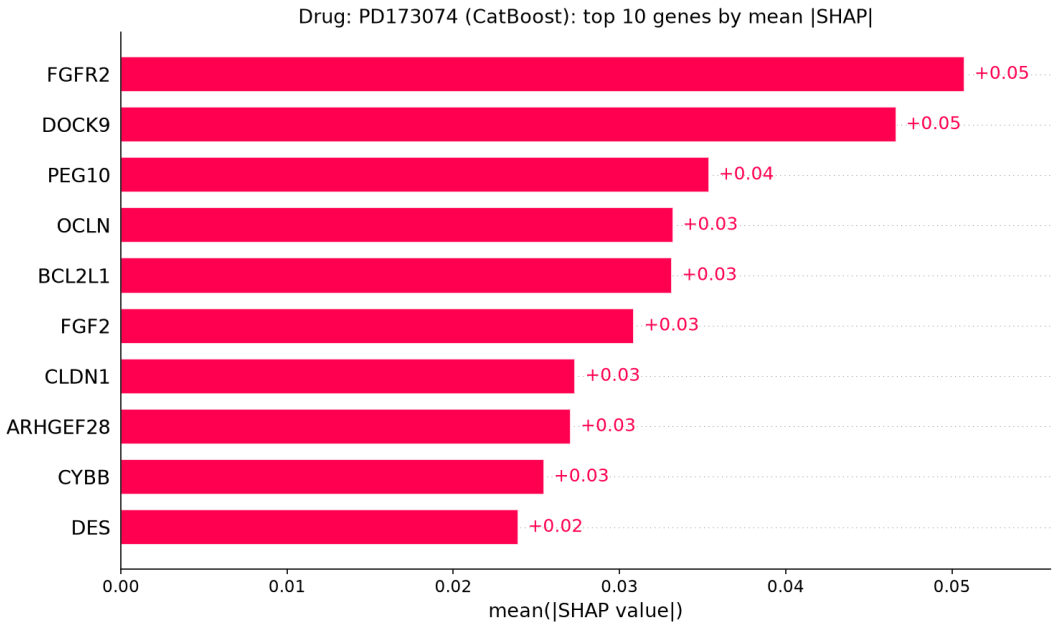


Figure 24. Top 10 genes by mean |SHAP| value for PD173074 (best model: CatBoost).

Description. Figure 24 shows the SHAP importance of the ten most influential genes for the CatBoost model of PD173074. The three most important genes are FGFR2 (0.051), DOCK9 (0.047), PEG10 (0.035); the tenth gene, DES, has a mean |SHAP| of 0.024.

Observations.

- FGFR2, the target of the FGFR inhibitor PD173074, is the top gene, and its ligand FGF2 is ranked 6th.
- The SHAP values are small and close together (0.051 for FGFR2 vs 0.047 for DOCK9). FGFR dependence is rare among cancer cell lines, so the strong signal comes from only a few lines, which also explains the low R^2 of 0.28.

Drug: Palbociclib

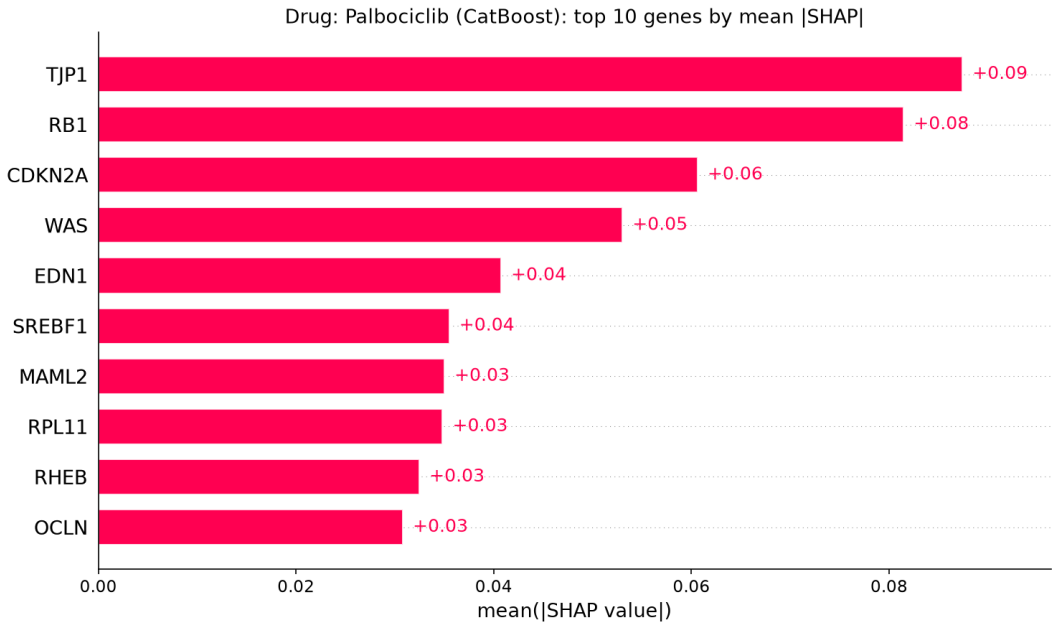


Figure 25. Top 10 genes by mean |SHAP| value for Palbociclib (best model: CatBoost).

Description. Figure 25 shows the SHAP importance of the ten most influential genes for the CatBoost model of Palbociclib. The three most important genes are TJP1 (0.087), RB1 (0.081), CDKN2A (0.061); the tenth gene, OCLN, has a mean |SHAP| of 0.031.

Observations.

- RB1 (2nd) and CDKN2A (3rd) are central to the drug's mechanism. Palbociclib blocks CDK4/6, which acts by phosphorylating the RB1 protein, so cells need functional RB1 to respond. CDKN2A encodes p16, the natural inhibitor of CDK4/6.
- TJP1 is the top gene (0.087), and the remaining genes have similar, moderate importance.

Drug: Pictilisib

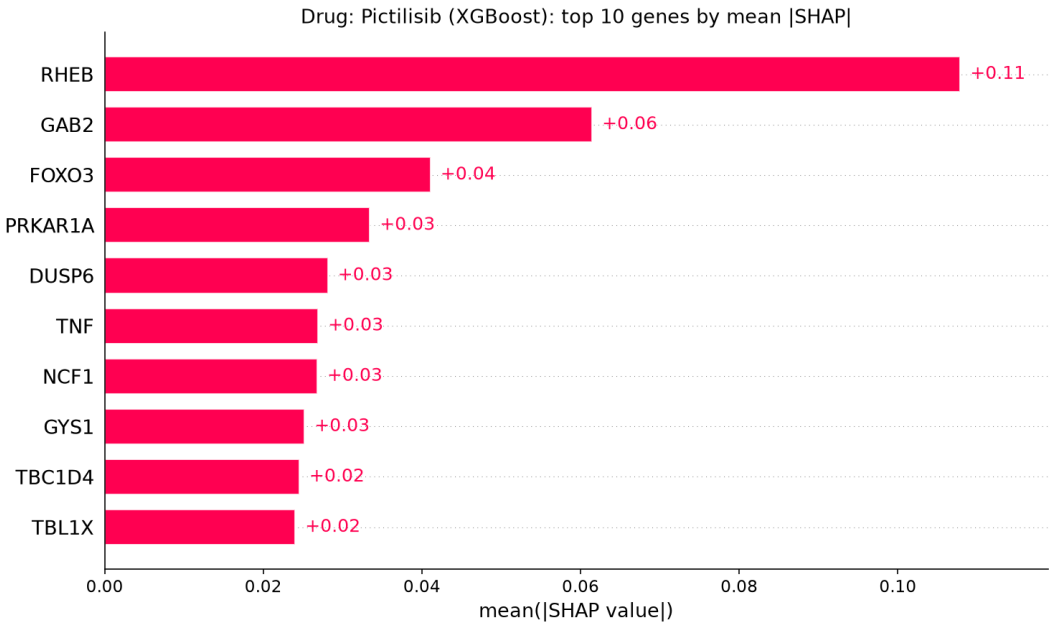


Figure 26. Top 10 genes by mean |SHAP| value for Pictilisib (best model: XGBoost).

Description. Figure 26 shows the SHAP importance of the ten most influential genes for the XGBoost model of Pictilisib. The three most important genes are RHEB (0.108), GAB2 (0.061), FOXO3 (0.041); the tenth gene, TBL1X, has a mean |SHAP| of 0.024.

Observations.

- Pictilisib is a PI3K inhibitor, and several top genes belong to the PI3K/AKT/mTOR pathway: RHEB (the top gene, 0.108) activates mTORC1, GAB2 activates PI3K, and FOXO3 and TBC1D4 are AKT substrates.
- Apart from RHEB, the SHAP values are small, consistent with Pictilisib being the hardest drug to predict (R^2 0.15).

Drug: Staurosporine

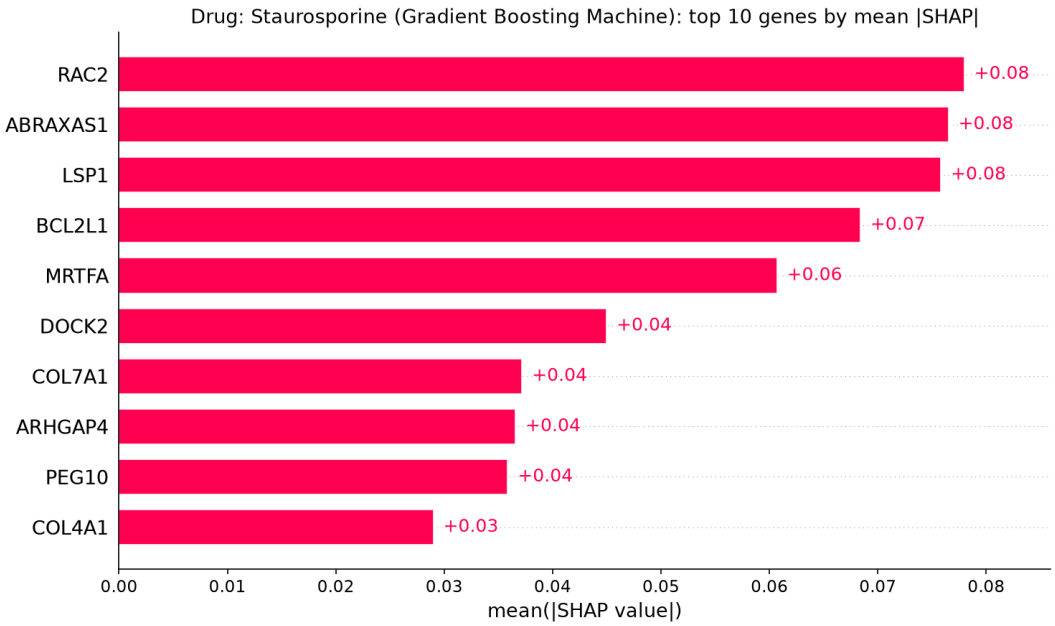


Figure 27. Top 10 genes by mean |SHAP| value for Staurosporine (best model: Gradient Boosting Machine).

Description. Figure 27 shows the SHAP importance of the ten most influential genes for the Gradient Boosting Machine model of Staurosporine. The three most important genes are RAC2 (0.078), ABRAXAS1 (0.077), LSP1 (0.076); the tenth gene, COL4A1, has a mean |SHAP| of 0.029.

Observations.

- RAC2, LSP1, DOCK2 and ARHGAP4 are expressed mainly in blood (hematopoietic) cells. The model therefore seems to learn cell type: blood-cancer cell lines respond differently to Staurosporine than solid-tumour lines.
- This makes sense for Staurosporine, which inhibits many kinases at once and has no single target gene. BCL2L1 (4th) again appears as a general survival gene.

Summary of Task 3

Table 17 lists the top 10 genes of every drug, most important first.

Table 17. Top 10 genes per drug by mean |SHAP| value.

Drug	Best model	Top 10 genes (most important first)
5-Fluorouracil	CatBoost	VEGFB, WAS, VAMP3, BCL2L1, EEF2, RHEB, RPS27L, MYBL1, LYN, SOCS2
Camptothecin	XGBoost	NQO1, TJP1, AGAP2, PHB2, HDAC11, THBS1, DLG4, APTX, TPT1, CTSD
Docetaxel	CatBoost	BCL2L1, BID, ABCB1, DFFB, APTX, ATP6AP2, PTPN1, RPTOR, PTMA, BTC
MG-132	CatBoost	CES1, CDC37, TINF2, S1PR2, ASAP1, FZD4, APTX, GATAD2A, KRT8, IL12A
MK-2206	LightGBM	FOXO3, KIFC3, DKK1, IRS2, IL13RA1, NCKAP1, MET, TH, SMURF2, GAB2
Navitoclax	XGBoost	BCL2, MCL1, YAP1, PMAIP1, MACROH2A1, MYB, MYOF, WNT3, APPBP2, CTNNA1
Nutlin-3a (-)	LightGBM	MDM2, RPS27L, AGAP2, MET, CDKN2A, BAX, CDKN1A, BCL2L1, KRT18, DDB2
Olaparib	Gradient Boosting Machine	MYOF, ITGA3, SDC4, TP53, BCL2L1, MET, LMNA, CTSD, HDAC11, TCF4
PD0325901	LightGBM	ETV4, DUSP6, SPRY2, GNA15, PLA2G4A, TLR4, FCGR2A, RIN1, FZD1, ARAP3
PD173074	CatBoost	FGFR2, DOCK9, PEG10, OCLN, BCL2L1, FGF2, CLDN1, ARHGEF28, CYBB, DES
Palbociclib	CatBoost	TJP1, RB1, CDKN2A, WAS, EDN1, SREBF1, MAML2, RPL11, RHEB, OCLN
Pictilisib	XGBoost	RHEB, GAB2, FOXO3, PRKAR1A, DUSP6, TNF, NCF1, GYS1, TBC1D4, TBL1X
Staurosporine	Gradient Boosting Machine	RAC2, ABRAXAS1, LSP1, BCL2L1, MRTFA, DOCK2, COL7A1, ARHGAP4, PEG10, COL4A1

Description. Table 17 summarises Figures 15-27 by listing each drug's ten most important genes in order of mean |SHAP| value.

Observations.

- For many drugs, the top genes are the drug's known target or markers of its pathway. This suggests the models learned real biology rather than noise: BCL2 for Navitoclax, MDM2 for Nutlin-3a, FGFR2 for PD173074, RB1 and CDKN2A for Palbociclib, ETV4/DUSP6/SPRY2 (MEK/ERK activity) for PD0325901, and AKT-pathway genes (FOXO3, RHEB, GAB2) for MK-2206 and Pictilisib.
- ABCB1, the drug efflux pump behind taxane resistance, appears for Docetaxel. MCL1, a known resistance factor to BCL2 inhibitors, is the second gene for Navitoclax.
- BCL2L1, a general cell-survival gene, is in the top 10 for 6 drugs (5-Fluorouracil, Docetaxel, Nutlin-3a, Olaparib, PD173074, Staurosporine). This suggests that the ability to avoid apoptosis

affects the response to many drugs.

- The size of the SHAP values follows predictability. The best-predicted drugs (Nutlin-3a, Navitoclax, PD0325901) have large top-gene values (0.19-0.27), while the hardest (MG-132, Pictilisib, PD173074) have small, flat values.
- For Staurosporine, a non-selective kinase inhibitor, the top genes are mostly blood-cell markers. When a drug has no single target, the model seems to rely on cell type.

Conclusion. SHAP turns the black-box boosting models into interpretable gene rankings. These rankings largely agree with known drug mechanisms, and they suggest candidate biomarkers for the less well understood drugs.

5. Conclusion

This assignment built drug-specific models that predict cancer cell line sensitivity (LN_IC50) to 13 drugs from 2,528 gene expression features, and explained the best models with SHAP.

- **Data (Task 1).** The dataset is balanced, with 961-963 cell lines per drug, so all drug-specific models are trained on the same amount of data.
- **Regression (Task 2).** Six tree-based regressors were compared with 5-fold cross-validation on MAE, MSE, RMSE and R^2. Gradient boosting methods are best for all 13 drugs (CatBoost 5, XGBoost 3, LightGBM 3, GBM 2). They are often nearly tied with each other, but they clearly beat Random Forest and, especially, the single Decision Tree, which overfits the high-dimensional data. The best R^2 ranges from 0.15 (Pictilisib) to 0.60 (Nutlin-3a).
- **Explainability (Task 3).** SHAP identified the top 10 genes for each drug. For the targeted drugs, these genes include the drug's own target (BCL2, MDM2, FGFR2) or well-known pathway markers (ETV4, DUSP6, RB1, FOXO3), which supports the validity of the models.

Limitations and future work. Gene expression explains only part of the variation in drug response. Adding other data types, such as mutations, copy-number changes and tissue of origin, could improve the models. The hyperparameters were set by hand. Tuning them with nested cross-validation, or selecting features before training, could further improve accuracy and make the SHAP rankings more stable.